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TA-RAG: Confidence-Calibrated Statement-Level Verification for Hallucination Resilient Regulatory Knowledge Systems

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ABSTRACT

Despite the heavily deployment of large language model (LLM) integrated with Retrieval Augmented Generation (RAG) for enterprise's regulatory systems, they still generate hallucinated responses unsupported by the retrieved information. In highstakes areas such as banking, legal compliances, this can present potentially real liability and operational risks. In this paper, we propose a Trust Aware RAG Framework and demonstrate that it can be used to transparently incorporate multistage statement level verification and confidence scoring in the standard RAG pipeline. Generated answers are effectively decomposed into atomic claims and determined to be correct or not with respect to retrieved documents according to a support-based confidence score, achieving calibrated answer rejection with poor evidence coverage. Tested on a real-world RBI regulatory document corpus with factual and outofscope queries, the system shows significant reduction in hallucinations and increased reliability over RAG baselines. The proposed methodology provides a practical way to a more secure and reliable enterprise knowledge system.

Keywords: Retrieval Augmented Generation (RAG), LLM, Hallucination Detection, Trust Aware AI, Regulatory Knowledge Systems, Confidence Scoring, Statement Level Verification, Enterprise AI, Answer Rejection mechanism.

1. INTRODUCTION

The Retrieval Augmented Generation (RAG) has become the go to approach to deploy Large Language Models (LLMs) in enterprise knowledge and regulatory systems, as it contextualizes responses with external documents using semantic retrieval and context incorporation. However, RAG also have hallucinations. The hallucinations are model generated information that is not supported or incorrect based on the provided evidence. In high impact fields like legal or banking compliance, even the slightest unsupported output can spread misinformation and result in huge legal, financial or operational risks. The need for provable grounding and calibrated uncertainty handling is thus crucial in the deployment of AI systems at an enterprise level [2].

There is a need for something which checks the trust worthiness of sources and its statements while reasoning with it. In this paper, we propose a Trust Aware RAG where reliability will be increased by bringing in statement level verification and confidence-based answer rejection. It breaks generated responses down into atomic claims, checks their truth with retrieved documents and eventually calculates support-based confidence score that reflects where answer should be accepted or rejected. Empirical evaluations on an RBI regulatory corpus show that the introduced model significantly mitigates recoverable hallucinations and increases the faithfulness compared to its baseline RAG models. Proposed framework has evaluated using structured query pattern and well-defined matrices that make system reliable.

Along with explicit verification and withholding in the generation pipeline, this framework extends retrieval only grounding to safer and more evidence aware enterprise AI deployments. The proposed architecture takes a verification-

based view and is differentiated by the focus on fact embedded, constrained uncertainty and robust decision support within the regulatory knowledge system.

2. LITERATURE REVIEW

From overall progress in LLM, RAG many researchers studied phenomenon of hallucination in large language models (LLMs) with special focus on retrieval augmented generation (RAG) systems. Kong et al. proposed a multiple respective consistency checking framework in black box and zero resource setting, where the detection of hallucination is based on the intersection of confidence estimated from multiple randomly sampled response sets and queries reconstructed to mitigate the overconfidence stemming from single perspective self-evaluation approaches, yielding better detection performance across datasets without requiring extra expertise or supervision. External knowledge or internal model, although they concentrate in detection but not its mitigation inside complete RAG pipelines [1].

In addition to detection-based methods, Tikka et al. explored hallucination prevention in three industrial deployment setups and revealed that system level interventions like fast guardrails, output decoration, temperature control, model modularization and retrieval gating effectively mitigate hallucinations without compromising safety. On the other hand, these solutions rely heavily on constrained domain workflows with no guarantee of generalizing to open domain settings for RAG [2]. At more conceptual level, Zhang had a broader survey on hallucination in retrieval augmented LLMs with the classifications of hallucination cause. They identify failure to retrieve information from an unreliable source or under ambiguity, generation deficit such as context conflict, limit of the model capability, their work focusing on prompt engineering and post generation correction

strategies, but without suggesting any verification that can be seamlessly embedded into working RAG systems [3]. Besides retrieval optimization, the hybrid retrieval was further studied by Mala and closely related work by Gezici and Giannotti which utilized sparse and dense retrieval strategies along with query expansion and dynamically weighted rank fusion to attain better grounding, mitigate hallucination and rejection rates as well as increase answer accuracy [4]. However, their work is largely an empirical comparative study which suggests on looking at more advanced reranking schemes as well as cross domain generalization. Meanwhile, Elchafei and Abu Elkheir proposed a span level hallucination detection model combining SRL, textual entailment using DeBERTa models based on token level confidence scores to verify verbal question answering, while also specifying enhanced hallucination validation through GPT4 and Llama fact checking methods, however their approach is still limited in its reliability for morphologically rich languages and entailment robustness [5].

Similarly, Ajmal et al. studied the problem of hallucination in 500 structured question-answer pairs from fifteen disciplines and showed that postdoc retrieval verification can significantly reduce the frequency of hallucinations while enhancing accuracy, precision, recall and F1 scores; GPT4 is better than small models despite moderate limitations on recall analyzing them further shows the need for improvements in both detection and mitigation [6]. Wu et al. further introduced Multi-RAG, a knowledge enhanced RAG framework for mitigating hallucination in multisource retrieval by building multisource knowledge graphs and leveraging multi-level confidence scoring to filter untrustworthy information to achieve better retrieval reliability and factual consistency on the question

answering tasks with multichip reasoning and cross domain variety, while dynamic uncertainty estimation as well as adaptive hallucination rejection need further dedicated exploration [7]. Finally, Magesh et al. empirically evaluated RAGbased legal research tools, and demonstrated that although such methods can decrease hallucinated responses as compared to standalone models, humanlike error rates (17%33%) remain a challenge in professional legal contexts due to naive retrieval, erroneous reasoning and extraneous authority selection, yet their contribution is mainly on evaluation and failure mode rather than automated mitigation [8].

While many studies have addressed hallucination in RAG systems through single approaches such as detection, retrieval optimization or system-level safeguards. Still no system provides a complete, integrated solution. To address this gap, we propose a multi-stage retrieval pipeline combined with statement-level claim decomposition, evidence verification, confidence scoring, and calibrated rejection, ensuring reliable, grounded responses in enterprise and knowledge-critical settings.

3. METHODOLOGY

3.1 System Overview

This work proposes a TrustAware Retrieval Augmented Generation (TA-RAG) framework developed to mitigate hallucinations effects in regulatory question answering. The architecture Fig.1 extends a standard RAG pipeline by introducing structured evidence verification and a calibrated rejection mechanism. This ensures that generated responses remain grounded in retrieved regulatory documents. The key objective is not only to generate context aware answers but to explicitly quantify the evidential support behind each claim before presenting the final output.

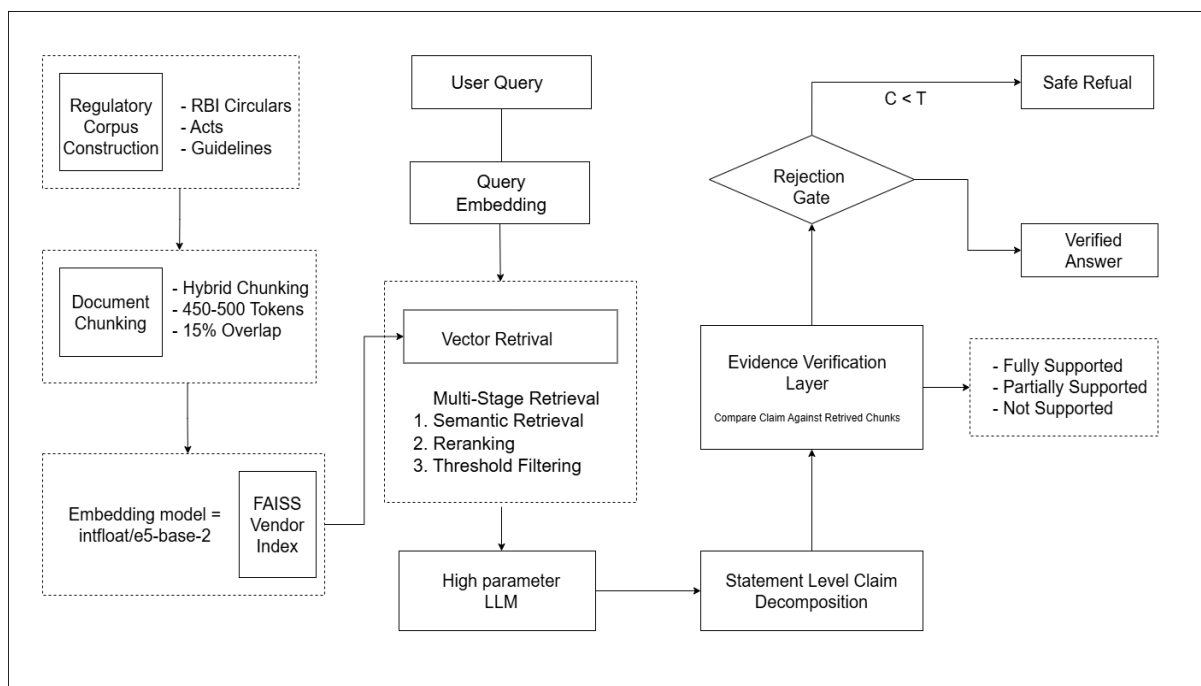


Fig.1 Architecture of the Proposed Trust Aware RAG (TA-RAG) Framework

3.2 Data collection

For this study, we collected regulatory corpus from official website of the Reserve Bank of India (RBI). The dataset consists of 6 regulatory documents,

including master circulars, governance guidelines, and banking framework documents relevant to supervisory and compliance operations. The corpus statistics are summarized in Table 1

Table 1: Data Statistics

Parameter	Approximate Value
Documents	6
Total pages	1 6 2
Raw text size	4.9 MB
Extracted text tokens	60,000 to 80,000

3.2.1 Chunking Strategy

As data collected has structured and hierarchical nature of regulatory text, hybrid semantic chunking technique was adopted. This specifically optimized for regulatory text. Each document was divided into semantic segments of 450 to 500 tokens with 15% overlap to preserve contextual continuity. This process created 112 chunks. It has metadata including source document, section, page and year. This structured segmentation supports

precise retrieval and minimizes context fragmentation.

3.2.2 Embedding and Vector Indexing

All chunks were converted into semantic embeddings using “intfloat/e5-base-2” model with 768-dimensional vectors. The vectors were indexed in a FAISS vector store for similarity search. The system retrieves the top 5 to 8 relevant chunks per query. The 112 vectors are stored for efficient semantic search across the regulatory corpus.

Due to computational constraints during experimentation, evaluation was conducted on the curated 6document corpus. However, the proposed architecture is scalable and can be extended to larger regulatory datasets without structural modification.

3.3 Baseline RAG Implementation

The baseline RAG pipeline embeds the query using “intfloat/e5-base-v2, retrieves topk relevant chunks from a FAISS index with “IndexFlatIP” and generates answers using an open-source instruction tuned LLM [3]. The retrieved chunks are concatenated as context without additional verification. While this enables semantic grounding, the system may still produce unsupported or overconfident claims due to the absence of explicit trust validation mechanism, motivating the need for trust aware verification mechanism.

3.4 Proposed Trust Aware RAG Framework

3.4.1 Multi Stage Retrieval

TA-RAG introduces a three-stage retrieval pipeline: initial semantic retrieval, cross encoder reranking for relevance scoring and similarity threshold filtering. This process reduces propagation of irrelevant evidence and improves retrieval precision.

3.4.2 Statement Level Claim Decomposition

Generated responses are decomposed into atomic factual claims. Each claim is independently verified against retrieved evidence, allowing fine grained assessment of answer validity.

3.4.3 Evidence Verification Layer

Claims are categorized as fully supported, partially supported or unsupported using semantic entailment testing. A support score for each claim is computed as:

$$S_i = \frac{\sum \text{Evidence support from retrieved chunks}}{\text{total retrieved chunks}}$$

where S_i is the support score of claim i .

3.5 Confidence Scoring mechanism

Answer level confidence is calculated as the average support score across all claims:

$$\text{Confidence} = \frac{1}{n} \sum_{i=1}^n S_i$$

where "n" is the number of claims.

Unsupported claims receive penalty and partial support reduces the normalized confidence score.

3.6 Calibrated Answer Rejection

To prevent hallucinated outputs, a rejection policy is implemented.

3.6.1 Rejection Conditions

An answer is rejected if:

- Confidence < threshold ($\tau = 0.65 - 0.75$ calibrated experimentally)
- Any critical claim has "S" _ "i" < 0.5
- Retrieved evidence coverage < 60%

3.6.2 Rejection Output

Instead of fabricating information, system returns:

“Insufficient verified evidence available in current regulatory corpus.”

This converts hallucination risk into safe refusal.

3.7 Model Configuration

The core configuration settings of the proposed TA-RAG framework present table 2, including LLM setup, context window size, decoding parameters and retrieval settings.

These parameters were selected to ensure stable generation, controlled variance and reliable evidence grounded response in regulatory question answering.

Table 2: Model Configuration Parameters

Query Type	Count
In scope factual	15
Multi hop regulatory	5
Out of scope	5
Ambiguous queries	5
Total	30

Table 3: Query Type

Metrics	Description
Claim HR	Fraction of unsupported claims among all claims.
Binary HR	Fraction of answers containing any unsupported claim.
SupportAccuracy	Claims supported by evidence
Correct Rejection	OOS queries correctly rejected

8. Evaluation

Evaluation of the system was conducted using a structured query approach [6], as summarized in Table 3. Four metrics were used to assess performance, which are

detailed in Table 4. This evaluation ensures an accurate and reliable measurement of the hallucination rate (HR).

Table 4: Metrics

Component	Implementation
LLM	Mistral7BInstructv0.2
Contextwindow	3000
Temperature	0.2(lowvariance)
Maxtokens	768
Retrievalk	5 to 8
Verification threshold	0.65

Through the methodology, the proposed Trust Aware RAG (TA-RAG) framework significantly improves over traditional retrieval augmented generation through multi stage retrieval, statement level verification, support weighted confidence scoring and calibrated rejection. The

experimental setup and evaluation framework are discussed in this section and presents a comprehensive comparison between RAG and trust aware architecture in next section.

4. RESULT AND DISCUSSION

4.1 Results

The performance of the proposed TA-RAG model was tested on different regulatory queries, as shown in Fig. 3. TA-RAG decreases claim hallucination rate from 20% to 8% binary hallucination

from 26% to 11%, which also improves claim support accuracy. Also correct rejection of out-of-scope queries from 5% to 90% (Fig. 2). These results highlight that statement level verification with calibrated rejection substantially enhances reliability over baseline RAG.

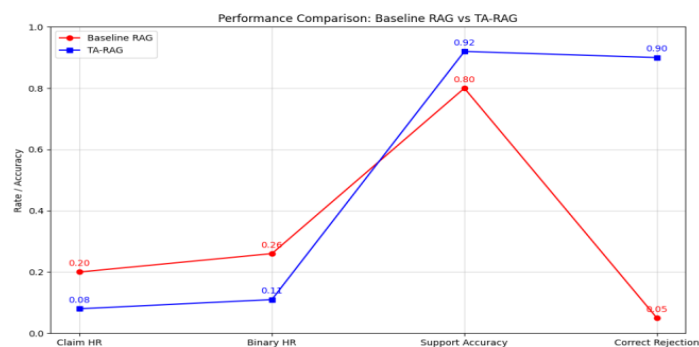


Fig. 2 Performance Comparison

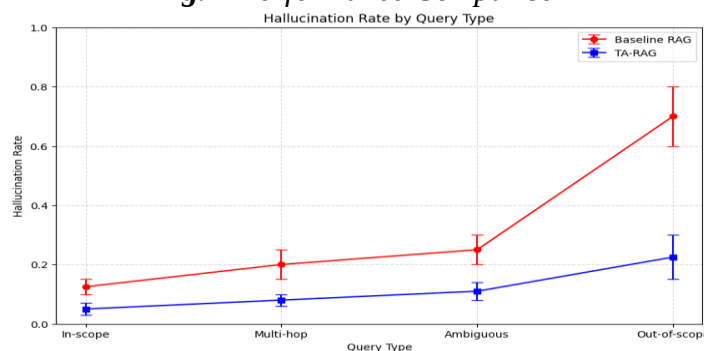


Fig. 3 Hallucination Rate by Query Type

4.2 Discussion

The results show that traditional RAG architectures are still inefficient to unsupported claim generation. It has a measured impact on regulatory scenarios where precision and fact-based reality matter. These vulnerabilities result from restrictive fetching relevance, lacking of filtered context information and the use of probabilistic language model that can propagate unsupported or fake partial statements. TA-RAG moves the system overhead of probabilistic answering to evidence informed decision support by combining claim decomposition, semantic validation and calibrated rejection. This disciplined structure is capable of fine-grained checking at content generation

time and has incorporated checkable confidence controls that assist in governing system output. The framework also shows that trust in regulatory AI systems should not be simply assumed at retrieval time but imposed through structured verification mechanisms. This verification inspired approach suitable for deploying trustworthy AI systems in highstakes enterprise context where correctness, openness and accountability are crucial.

5. CONCLUSION

In this paper, we have tried Trust Aware RAG architecture integrating statement level verification and calibrated confidence scoring to enhance factual

grounding on regulatory QA. The proposed TA-RAG is effective in mitigating hallucination and enhancing the accuracy of claim support and reliability for out-of-scope rejection, resulting in more alignment between generated outputs and verifiable evidence. By validating with evidence and refusing in control, the system turns traditional RAG into a verification pipeline for mission critical enterprise where trust is required. These findings underscore the need for structured verification to deploy trustful regulatory AI systems, as well as suggest integrating transparency, accountability and confidence aware decision-making processes in enterprise scale language model applications.

6. FUTURE SCOPE

In future additions, TA-RAG can be extended by introducing more powerful models and real time uncertainty estimation for better hallucination. In addition, integrating more domains and languages to the corpus would improve generalization of the model, as well as incorporating reinforcement learning and user feedback loops for lifelong system tuning and adaptive trust modeling. These advancements will enable the creation of resilient, domain independent trust aware AI systems that are able to operate reliably in complex, high stakes decision support contexts.

REFERENCES

1. L. Kong, X. Zhong, J. Chen et al., "Multi perspective consistency checking for large language model hallucination detection: A black box zero resource approach," *Frontiers of Information Technology & Electronic Engineering*, vol. 26, pp. 2298–309, 2025, doi: 10.1631/FITEE.2500.
2. P. Tikka, J. Karjalainen, A. Alesani, and V. Goriachev, "Large language model hallucination mitigation in three industrial use cases," *IEEE Access*, early access, 2026, doi: 10.1109/ACCESS.2026.3659997.
3. W. Zhang and J. Zhang, "Hallucination mitigation for retrieval augmented large language models: A review," *Mathematics*, vol. 13, no. 5, Art. no. 856, 2025, doi: 10.3390/math13050856
4. C. S. Mala, G. Gezici, and F. Giannotti, "Hybrid Retrieval for Hallucination Mitigation in Large Language Models: A Comparative Analysis," *arXiv preprint arXiv:2504.05324v1 [cs.IR]*, 2025, doi: 10.48550/arXiv.2504.05324.
5. P. Elchafei and M. Abu Elkheir, "Hallucination Detectives at SemEval 2025 Task 3: Span Level Hallucination Detection for LLM Generated Answers," in *Proc. 19th International Workshop on Semantic Evaluation (SemEval 2025)*, Vienna, Austria, Jul. 2025, pp. 601–606
6. R. H. Ajmal, M. U. Sarwar, M. K. Hanif, and M. I. Khan, "Evaluating the Effectiveness of Advanced Language Models in Detecting and Mitigating Hallucinations Using Structured Question Answering, Novel Metrics, and Post Hoc Retrieval," *IEEE Access*, vol. 13, pp. 173805–173812, 2025, doi: 10.1109/ACCESS.2025.3613851.
7. W. Wu, H. Wang, B. Li, P. Huang, X. Zhao and L. Liang, "MultiRAG: A Knowledge Guided Framework for Mitigating Hallucination in Multi Source Retrieval Augmented Generation," 2025 *IEEE 41st International Conference on Data Engineering (ICDE)*, Hong Kong, Hong Kong, 2025, pp. 3070–3083, doi: 10.1109/ICDE65448.2025.0030
8. Magesh, V., Surani, F., Dahl, M., Suzgun, M., Manning, C.D. and Ho, D.E. (2025), *Hallucination Free? Assessing the Reliability of Leading AI Legal Research Tools*. *J Empir Leg Stud*, 22: 216–242. <https://doi.org/10.1111/jels.12413>